

LZSCC.241 Coursework

Human-Computer Interaction Project

Prepared by: Amna Asif (a.asif2@lancaster.ac.uk)

Contents

LZSCC.241 Coursework Human-Computer Interaction Project.....	1
<i>Rules</i>	2
<i>Project Overview</i>	2
Submission Details	3
<i>Activity 1: Problem Definition and Conceptualization</i>	3
<i>Activity 2: Need-finding</i>	3
<i>Activity 3: Exploration of Design Alternatives</i>	3
<i>Section 4: User Experience UX Design Integration</i>	4
<i>Activity 5: Prototyping</i>	4
<i>Activity 6: Usability Evaluation</i>	4
<i>The contents of the portfolio</i>	5
LZSCC 241 Group Project Marks Rubric (25 Marks)	6
Group Members' Contribution Table (Example)	8
List of innovative project titles:	9
References and Resources	9
Example project portfolios	9
LZSCC 241. HCI Workshop Portfolios Rubric (5 Marks)	11

- ✓ *Course Instructor: Dr. Amna Asif*
- ✓ *Course Title: LZSCC. 241 Human-Computer Interaction Project*
- ✓ *Date Set: Week 3*
- ✓ *Submission Deadline Friday 12th December 2025, week 10 at 14:00*
- ✓ *Submission components: 1) Online project portfolio website link, 2) Individual portfolio link*
- ✓ *Coursework value: 30% (25% group work 5% individual portfolio)*
- ✓ *Type group project.*
- ✓ *Number of group members allowed 5-6 members.*

Rules

- You need to submit various activities, which may be submitted in PDF (presentation or document), writing style and URL (for Figma prototypes and GitHub webpage) and videos. Therefore, each project activity must contain balanced paragraphs with citations and referencing, captioned images and tables, and proper structuring.
- If you collect personal data from users, use an informed consent form.
- Each group should submit a group member contribution table in the report appendix.
- No extensions are allowed for project submission.
- In the case of plagiarism, we will follow Lancaster University's policy as <https://portal.lancaster.ac.uk/ask/download?document=/media/lancaster-university/content-assets/documents/student-based-services/asq/principles-policies-and-guidelines/Plagiarism-Framework.pdf>.
- This project must be completed as a group effort.
- All of the coursework submissions of LZSCC.241 follows AI Category Red for all written, prototype and video recording submissions; however, the personas' images follow the **GREEN** Gen AI category:

AI Category **RED**: The use of generative AI tools (such as ChatGPT or similar) is not permitted for the report in the assessment.

GREEN: Gen AI Tools have an integral role

Project Overview

The Human-Computer Interaction (HCI) course project is designed to enhance your practical understanding of interaction design. This project allows your team to address real-world problems, design solutions, and create a product prototype. You must choose a topic of interest to complete your project. The following are the objectives of the HCI course project:

- Define and present the problem effectively.
- Apply a user-centered design process.
- Practice various need-finding methods, requirements analysis, and requirements documentation.
- Develop a conceptual model.
- Create alternative low-fidelity (paper-based) and high-fidelity prototypes (using a prototyping tool).
- Evaluate the prototypes using different methods.
- Present evaluation results in a report.

Submission Details

Activity 1: Problem Definition and Conceptualization

The students must articulate and present the problem effectively. They may choose to articulate the by creating a concept video or visuals to state the problem. The problem must be conceptualised through textual, visual (figure), or video scenarios. Understanding the system's users is foundational in the user-centred design process. Consequently, the submission of this activity include:

1. Clearly define the problem through visuals or provide a concept video.
2. Use personas, scenarios, textual descriptions, figures, or videos to conceptualize the problem.

Activity 2: Need-finding

The process of requirements gathering or identifying needs can involve various methods that can be used individually or in combination. These methods include conducting interviews, organizing focus groups, performing diary studies, distributing survey questionnaires, making direct observations, studying relevant documentation, conducting contextual inquiries, and researching similar systems.

Subsequently, scenarios, task analysis, and use case diagrams will be used to analyse the collected requirements. You are required to submit in presentation format the following:

1. Data and information collected through at least **two** of the various requirements-gathering methods, such as interviews, focus groups, questionnaires, direct observations, or documentation review. Presentation of all the stages of the requirement gathering process.
2. A complete list of **functional, non-functional, and usability/user experience requirements**.
3. An analysis of the collected requirements using personas (at least one for each user), scenarios (each scenario highlighting problems faced by personas while using the system), task analysis (a detailed examination of all components within the project's scope), and use case diagrams.
4. Justification for the requirements extracted from the data collected using different requirements-gathering methods.

Activity 3: Exploration of Design Alternatives

The exploration of alternative design solutions for the system involves a range of activities. The three main activities employed to translate users' needs into the design are (1) understanding the users' tasks and context of use, (2) developing alternative designs using low-fidelity prototyping techniques, and (3) evaluating these design ideas.

You will need to develop storyboards to understand users' tasks and context of use, identify prototyping techniques, and organize brainstorming sessions (present at least three strong design alternatives and one intentionally poor design alternative). Translate conceptual design ideas into tangible representations (such as sketches, storyboards, or prototype cards), conduct formative evaluations of your design alternatives, and select the preferred design, making informed decisions about which alternative(s) to proceed with based on formative feedback results.

You can explore design alternatives by:

- **Generate** many ideas and variations
- **Reflect** on all ideas (formative evaluation)
- **Choose** the ones that look most promising
- **Develop** them in parallel

- **Add** new ideas as they come up
- *Then* **iterate** your final choice

Therefore, your submission in presentation format must include:

1. An understanding of tasks and context using storyboards.
2. The minimum required variations of low-fidelity prototypes, along with appropriate scope and resolution. Additionally, justify your choices regarding the scope and resolution of the prototypes.
3. A formative evaluation of your prototypes.
4. A description of the iterations that led to your final choice.

Section 4: User Experience UX Design Integration

In one of the workshops, you have been practising different UX design methods called design journey mapping. These consist of Empathy mapping, Customer journey mapping, Experience mapping, and Service blueprinting (<https://www.nngroup.com/articles/ux-mapping-cheat-sheet/>). The user experience can be enhanced through timeline or wheel methods. In this coursework, you are required to develop a design journey mapping using suitable approaches for your project.

Activity 5: Prototyping

During this phase, the emphasis is on developing high-fidelity prototypes. This involves utilising computer tools such as Figma and Software Development Kits (SDKs) to develop an interactive prototype of the proposed solution. This tangible prototype is a key stepping stone for further evaluation. To develop high-fidelity prototypes, consider the following:

1. In this phase, you are required to develop high-fidelity prototypes while considering the prototype's scope, whether horizontal or vertical. Justify your choice of scope.
2. When designing UI components, consider design implications, aspects of human vision, and the selection of UI components, such as colour palettes and styles, based on user needs, metaphors, error handling and following adequate design guidelines.
3. Justify your choice of high-fidelity prototypes based on the research and analysis conducted in the previous activities.

You must share the Hi-fidelity prototype link as well as a recorded maximum 2-minute demo to present how your prototype follows the conceptual design, usability guidelines, principles, and the research and design issues of the cognitive process of *attention, perception, and memory*.

Activity 6: Usability Evaluation

Your project concludes with an extensive evaluation phase, emphasising summative evaluation significantly. This phase is critical within the project. Students can choose between conducting usability testing or experimenting with real users, or alternatively, they can opt for expert evaluation, utilising methods such as heuristic evaluation or cognitive walkthrough, which involve seeking feedback and insights from field experts.

Adhering carefully to suitable evaluation methodologies ensures a rigorous and thorough assessment of the interactive prototype that has been developed. The evaluation process demands a good investment of time and effort.

Therefore, in this phase, you are required to report:

1. Evaluate the high-fidelity interface developed in the last phase.

2. Apply at least one of the evaluation methods that you learnt. And justify why particular evaluation methods are most suitable in the context of your project.
3. Report the goals of the evaluation, participants, apparatus, procedure, results (with adequate tables and graphs), results discussion and your reflection on usability results.

References and citations

You must include citations and references using APA, IEEE or Harvard style throughout the report. For example, when making a claim or discussing a specific technique, cite the material from a reliable source.

Appendix

This section should include the following information:

1. Group Members' Contributions: Provide details of each group member's contribution to the project. Include a table summarizing each member's specific roles and tasks.
2. Data Sources: Include relevant data, such as interview transcriptions, questionnaires, usability testing data, etc.

The contents of the portfolio

- Activity 1: Problem Definition
- Activity 2: Need-Finding
- Activity 3: Exploration of Design Alternatives
- Activity 4: User Experience Design
- Activity 5: Prototyping
- Activity 6: Usability Evaluation
- References

LZSCC 241 Group Project Marks Rubric (25 Marks)

	Exemplary (100-80 percentage)	Proficient (79-60 percentage):	Basic (59-40 percentage):	Below Expectations (39-0 percentage):
Activity 1: Problem Definition and Conceptualization (10%)	Clear, concise, and well-articulated problem statement or concept video. Effectively uses scenarios (text, visuals, or videos) to conceptualize the problem. Creates detailed user personas that provide strong insights into the target users and their needs.	Adequate problem statement or concept video with some clarity. Uses scenarios to conceptualize the problem but may lack depth or creativity. User personas are present but may lack detail or specificity.	Problem statement or concept video is present but lacks clarity. Limited use of scenarios to conceptualize the problem. User personas are vague or do not fully represent user needs.	Problem statement or concept video is missing or unclear. Scenarios are poorly developed or absent. No user personas or personas lack any relevant information about users.
Activity 2: Need-Finding (20%)	Thorough use of at least two methods (e.g., interviews, focus groups, or observations). Requirements are clearly listed, covering functional, non-functional, and UX needs comprehensively. Analysis using scenarios, task analysis, and use case diagrams is insightful and demonstrates a clear understanding.	Uses at least two need-finding methods with adequate detail. Lists functional and non-functional requirements, though some may lack depth. Provides basic analysis using scenarios, task analysis, or use case diagrams.	Limited use of need-finding methods, or only one method is used in depth. Requirements listing is incomplete or lacks clarity. Minimal analysis provided, with limited use of scenarios or task analysis.	Need-finding methods are absent or poorly executed. Requirements listing is minimal or absent. Analysis is missing or superficial, with little to no use of scenarios or task analysis.
Activity 3: Exploration of Design Alternatives (20%)	Strong exploration of at least three good design alternatives and one poor alternative. Low-fidelity prototypes (e.g., sketches, storyboards) are well-crafted and demonstrate creativity and understanding. Formative evaluation is performed thoroughly, with clear justification for the preferred design based on feedback.	Adequate exploration of design alternatives, including at least two good and one poor option. Low-fidelity prototypes are present but may lack refinement. A formative evaluation is conducted, with some explanations for the chosen design alternative.	Limited design alternatives were explored, with minimal distinction between good and poor options. Low-fidelity prototypes are basic and lack detail. Formative evaluation is incomplete, and justification for the preferred design is unclear.	Few or no design alternatives are presented. Low-fidelity prototypes are missing or poorly executed. Formative evaluation is absent, with no justification for design decisions.
Activity 4: User Experience (UX) Design Integration (5%)	Comprehensive UX design journey mapping, using at least two methods (e.g., empathy	Adequate UX design journey mapping using one or two methods, with some detail. Alignment	Limited or superficial use of UX design journey mapping methods, with minimal alignment	UX design journey mapping is absent or poorly executed, with no alignment with

	mapping, customer journey mapping). Demonstrates strong alignment with project goals and user needs.	with project goals and user needs is evident but may lack depth.	with project goals. Lacks comprehensive insight into user needs.	project goals or user needs.
Activity 5: High-fidelity Prototyping (20%)	A high-fidelity prototype was created using tools like Figma or SDKs, showing interactivity and clear design implications, style and colour choice. The prototype is polished, engaging, and aligns well with the project objectives, with a complete scope and appropriate resolution.	The high-fidelity prototype is functional but may lack some polish or interactivity. Shows alignment with project goals but could benefit from additional refinement with partial scope and resolution.	Basic high-fidelity prototype with limited interactivity. The design may not fully align with project goals or may lack refinement with incomplete scope and resolution.	High-fidelity prototype is incomplete, lacks interactivity, or shows minimal alignment with project objectives.
Activity 6: Usability Evaluation (25%)	Evaluation is thorough, using feedback effectively to assess and refine the prototype. Demonstrates critical analysis. Also, complete, correct and structured reporting using adequate tables and visualisation.	Adequate use of the evaluation methods with some depth. Uses feedback for assessment but may lack thorough analysis or critical insights. Reporting is clear but could be more structured or detailed.	Limited use of evaluation methods, or only one complete method is used. Minimal analysis or use of feedback for improvement. Reporting includes some structure or visuals but is incomplete or inconsistent.	Evaluation is absent or poorly executed, with little to no critical analysis or feedback implementation. Reporting is unclear, unstructured, or missing essential elements.

Group Members' Contribution Table (Example)

Present this contribution table in the report appendix with team member contributions.

	Activity 1	Activity 2	Activity 3	Activity 4	Activity 5	Activity 6
Group member 1 (ID)	Prepared the scenarios	Conducted interviews. List down the functional requirements. Documentation. Developed the personas	Developed 1 good design idea. Developed the storyboard	Explored the UX techniques	Conducted the research	Planned the evaluations
Group member 2 (ID)	Recorded the video	Integrated the information in the portfolio. Prepared the personas	Prepare the presentation. Evaluated the design	Developed UX design 1	Developed the hi-fi prototype	Documentation
Group member 3 (ID)	Documentation	Identified usability requirements. Conducted the task analysis.	Developed 2 good design ideas	Developed UX design 2	Documentation	Conducted the evaluation 1
Group member 4 (ID)	Research the resources	Requirements collection using surveys. Developed the use case diagrams	Developed the bad design idea. Evaluated the good design	Completed the visuals.	Developed the hi-fi prototype	Conducted the evaluation 2
Group member 5 (ID)	Research the resources	Planned interviews.	Developed 2 bad design ideas	Completed the presentation.	Completed 2 screens.	Analysed the data

List of innovative project titles:

- HR system for Lancaster University Leipzig
- Language Learning Companion
- Smart Home Control Hub
- Inclusive Mobile Gaming
- Social Media Enhancement
- Fitness and Wellness Dashboard
- Wearable Technology App
- Device Security Optimization
- Emergency Response Visualization
- Community Engagement Platform
- Sustainable Transportation
- Voice Assistant Improvement
- Interactive Family Photo Albums
- Educational Game Accessibility
- Agricultural Efficiency
- Tourist Exploration App
- Human-Centric AI Assistant
- Human-Robot Interaction
- Collaborative Workspace
- Transportation Network Optimization
- Event Discovery and Management
- Healthcare Accessibility
- LU Leipzig apps
- Any other topic

References and Resources

1. Sharp, Helen, et al. Interaction design: beyond human-Computer interaction. Wiley, 2016.
2. <https://www.figma.com/>
3. https://balsamiq.com/wireframes/?gclid=Cj0KCQjwteOaBhDuARIsADBqRehKwbf60RPN-YnniEXy6P4T3cOmMsdpDAD-_Xk6YOPnTsUrc7lnqVkaAlxgEALw_wcB
4. <https://www.invisionapp.com/cloud/prototype>
5. <https://pencil.evolus.vn/>
6. <https://proto.io/>
7. <https://www.adobe.com/products/xd.html>
8. <https://marvelapp.com/>

Example project portfolios

<https://web.stanford.edu/class/cs147/projects24/Designing-for-Movement/ConcertBuds/>

<https://web.stanford.edu/class/cs147/projects24/Designing-for-Movement/PISSION/>

<https://web.stanford.edu/class/cs147/projects24/Job-Accessibility-Career-Development/HireReel/>

<https://web.stanford.edu/class/cs147/projects24/Job-Accessibility-Career-Development/UniVerse/>

<https://web.stanford.edu/class/cs147/projects24/Transforming-Language-Learning-AI/Mora/>

LZSCC 241. HCI Workshop Portfolios Rubric (5 Marks)

Criteria	Excellent (80–100%)	Good (79–60%)	Satisfactory (40–59%)	Poor (0–39%)
Completeness and Consistency (50)	All weekly activities are fully completed and regularly uploaded to GitHub. Portfolio shows consistent progress, clear organisation, and continuous improvement across weeks.	Most activities were completed with minor gaps or delays. Portfolio shows good engagement and progression but may lack full consistency or detailed iteration.	Several activities are missing or incomplete. Updates are irregular, with limited evidence of weekly engagement or progression.	Many activities are absent or poorly uploaded. The portfolio lacks structure and shows little engagement or progress throughout the module.
Design Thinking and HCI Application (30)	Demonstrates strong understanding and creative application of HCI principles. Each activity reflects user-centred thinking, design reasoning, and iteration based on feedback. Shows critical reflection and innovation in design solutions.	Applies HCI concepts appropriately with some design reasoning and reflection. Some activities show thoughtful user focus, but not always consistently.	Basic or partial application of HCI principles. Limited evidence of user consideration or iteration. Reflections are minimal or descriptive.	Little or no evidence of HCI understanding. Activities show no user focus, reasoning, or iteration.
Presentation (20)	Portfolio is well-structured, visually coherent, and professionally presented. The portfolio on GitHub is easy to navigate with help, labelled menus/icons, visuals, and documentation. Presentation reflects care and attention to detail.	Portfolio is clear and usable, though some inconsistencies exist in structure, visuals, or documentation. Shows good effort but lacks polish.	Portfolio is partly organised but lacks clarity or coherence. Minimal visuals or documentation reduces professional appearance.	Portfolio is disorganised, unclear, or incomplete. Poor presentation makes navigation and evaluation difficult.